Documentation

Image Compression Web Application

Abstract

The Image Compression Web Application is an innovative and interactive tool designed to facilitate the exploration and application of three advanced image compression techniques: Principal Component Analysis (PCA), Singular Value Decomposition (SVD), and Independent Component Analysis (ICA). This web-based platform allows users to upload images, select a compression technique, adjust the number of components for compression, and download the compressed image, all through a user-friendly interface.

The application aims to serve both educational and practical purposes by providing a hands-on approach to understanding the nuances of image compression. It offers a comparative analysis of the original and compressed images by displaying them side by side and calculating the Peak Signal-to-Noise Ratio (PSNR), a critical metric for evaluating compression quality. This enables users to observe the trade-offs between compression ratio and image quality in real-time.

The backend of the application is powered by Flask, a lightweight Python web framework, and integrates several powerful Python libraries including Pillow for image processing, NumPy for numerical computations, and scikit-learn for implementing the PCA, SVD, and ICA algorithms. This combination of technologies ensures a robust and efficient performance of the application.

Acknowledgement

I would like to extend my appreciation to my professors and academic mentors at **Kirti .M. Doongursee College**, for their support and for equipping me with the foundational knowledge, skills and guidance that helped me in the preparation of this project.

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1. Introduction

1.1. Overview

The Image Compression Web Application is a versatile and user-friendly platform designed to enable users to compress images using three distinct and powerful techniques: Principal Component Analysis (PCA), Singular Value Decomposition (SVD), and Independent Component Analysis (ICA). This web-based tool not only facilitates the compression of images but also provides a comparative analysis of the original and compressed images through the calculation of the Peak Signal-to-Noise Ratio (PSNR).

1.2. Purpose

The primary goal of this application is to provide an intuitive interface for users to experiment with different image compression techniques and understand their impact on image quality. By offering multiple compression methods and adjustable parameters, the application serves both educational and practical purposes, making it suitable for students, researchers, and professionals in the field of image processing and data compression.

1.3. Compression Techniques

- **Principal Component Analysis (PCA)**

  PCA is a statistical method used to reduce the dimensionality of data while retaining most of the variation present in the dataset. In the context of image compression, PCA transforms the image data into a set of orthogonal (uncorrelated) components, allowing for the reduction of redundant information and efficient storage.

- **Singular Value Decomposition (SVD)**

  SVD is a mathematical technique that factorizes a matrix into three other matrices, highlighting its intrinsic properties. When applied to images, SVD can decompose the image into singular values and vectors, enabling the reduction of less significant values and thus compressing the image. SVD is particularly effective for preserving essential details and structures within an image.

- **Independent Component Analysis (ICA)**

  ICA is a computational method for separating a multivariate signal into additive, independent components. Unlike PCA, which maximizes variance, ICA focuses on statistical independence, making it useful for applications where the underlying sources are assumed to be non-Gaussian and independent. In image compression, ICA can uncover latent structures within the image data, aiding in efficient representation.

1.4. Technical Details

The application is built using Flask, a lightweight WSGI web application framework for Python. It leverages several Python libraries for image processing and numerical computations:

- **Flask**: For creating the web interface and handling HTTP requests.
- **Pillow**: For image manipulation and processing.
- **NumPy**: For numerical operations and handling image data as arrays.
- **scikit-learn**: For implementing PCA, SVD, and ICA algorithms.

2. Features & Benefits

2.1. Key Features

- **Multi-format Image Upload**: Users can upload images in various formats such as JPEG, PNG, and BMP.
- **Technique Selection**: Users can choose between PCA, SVD, and ICA for image compression.
- **Adjustable Components**: Users can dynamically adjust the number of components used in the compression process to see how it affects the image quality and compression ratio.
- **Side-by-side Image Comparison**: The application displays the original and compressed images next to each other for easy comparison.
- **Download Option**: Users can download the compressed image for further use or analysis.
- **Quality Assessment**: The application calculates and displays the PSNR value, providing a quantitative measure of the compression quality.

2.2. Benefits

- **Educational Tool**: Ideal for learning and teaching image compression techniques and their effects.
- **Research Utility**: Useful for researchers experimenting with different image compression methods.
- **Practical Application**: Helps in reducing image file sizes for storage and transmission while maintaining acceptable image quality.

3. Project Structure

```
Project/
│
├──── app.py
│
├──── templates/
│      │
│      ├── select_technique.html
│      │
│      └── index.html
│
└──── static/
       │
       └── background.jpg
```

- **'app.py' :** The main Flask application file that handles routes, image compression logic, and PSNR calculation.
- **'templates/'** : Contains HTML templates for the web pages.
- **'static/'** : Contains static files like background images.

4. Setup Instructions

4.1. Prerequisites

- Python 3.x
- Python libraries:
    1. Flask
    2. Pillow
    3. NumPy
    4. scikit-learn

4.2. Running the Application

- Navigate the file where project is stored using terminal.
- Run the Flask application:

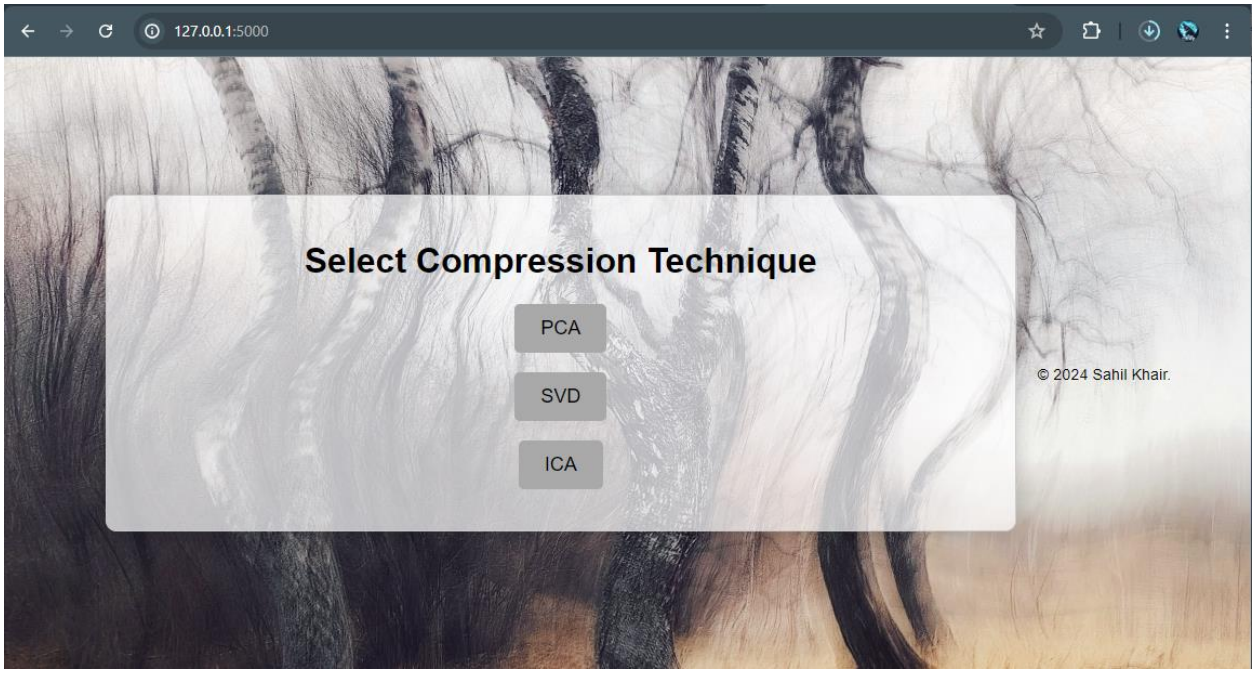
```sh
python app.py
```

- Open your web browser and navigate to http://127.0.0.1:5000/.

5. Usage

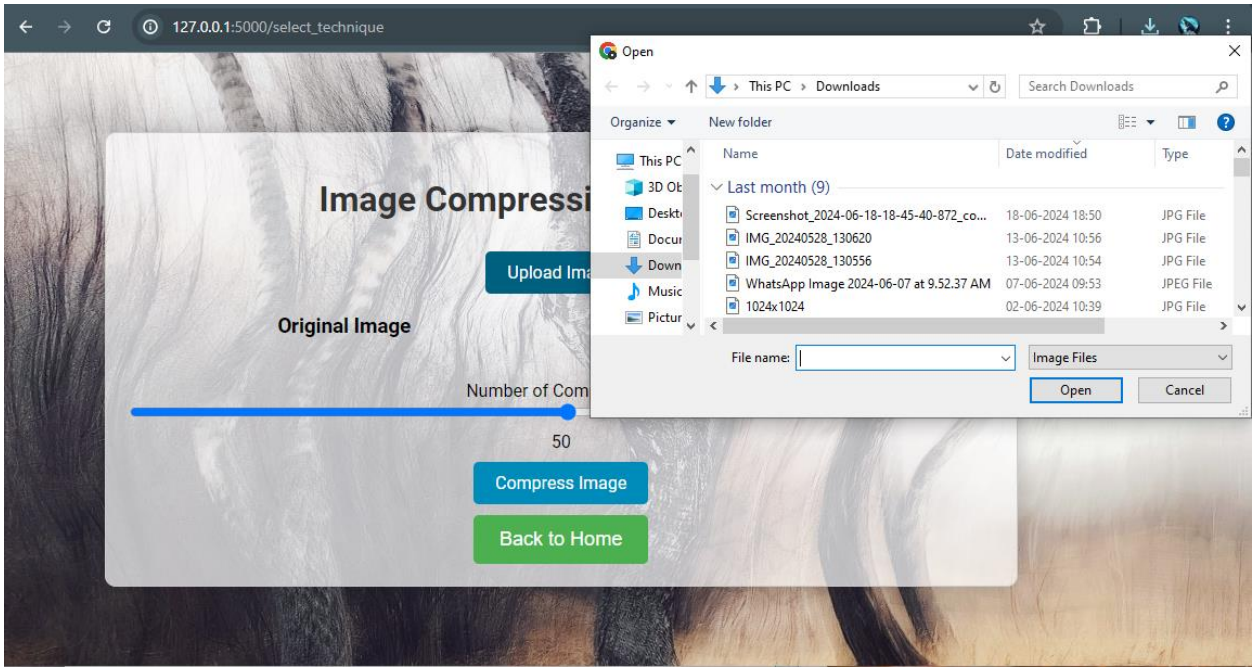
5.1. Technique Selection

- Open the application in your web browser.
- On the ‘select_technique.html’ page, select a compression technique by clicking on the PCA, SVD, or ICA button.



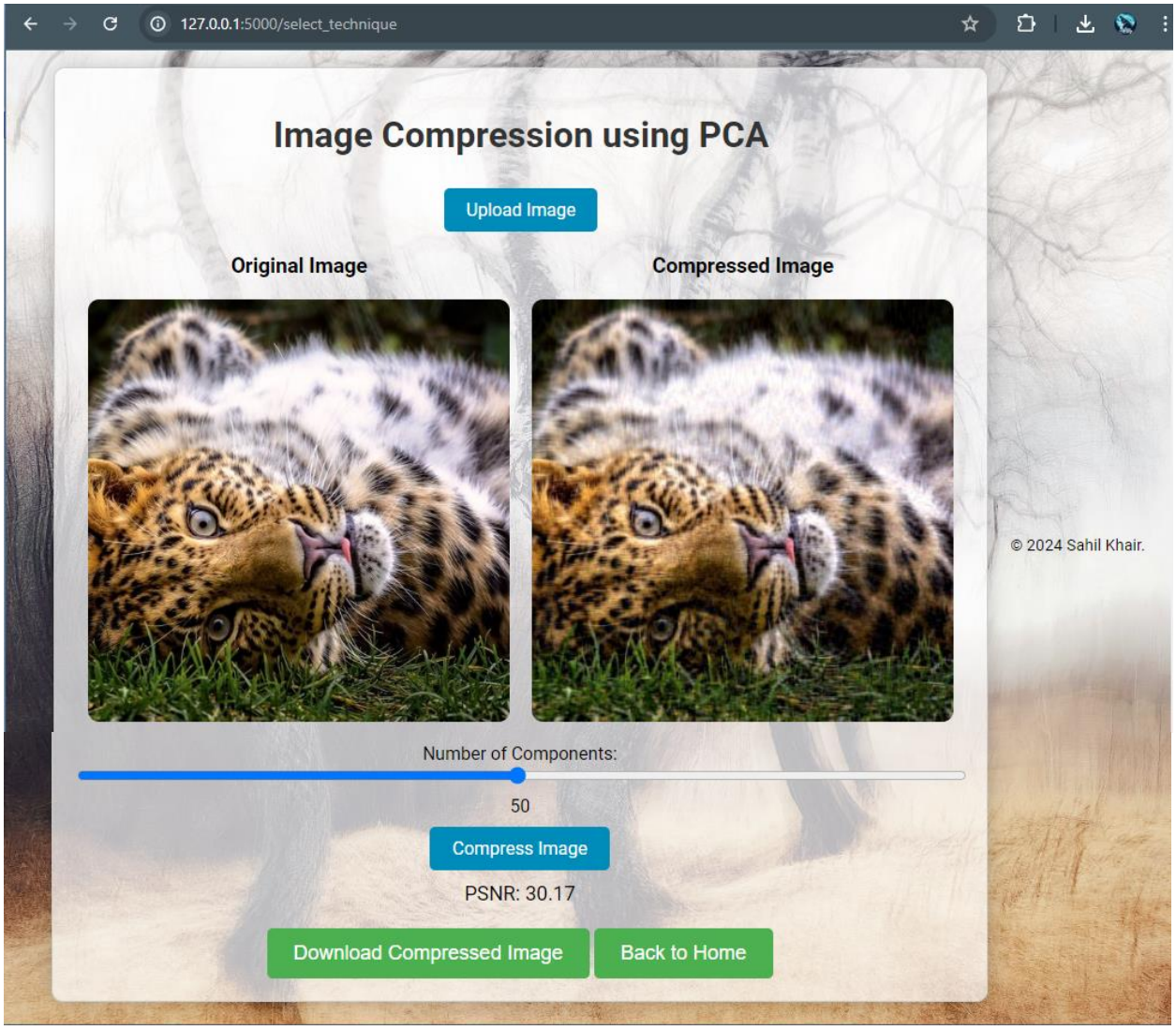
5.2. Image Upload and Compression

- On the ‘index.html’ page:
    - Click the "Upload Image" button to select an image file from your computer.
    - Adjust the number of components using the slider.
    - Click the "Compress Image" button to start the compression process.



5.3. Viewing Results

- The original and compressed images will be displayed side by side.
- The PSNR value will be shown below the images.
- You can download the compressed image using the provided download link.



6. Detailed Description

6.1. Flask Backend (‘app.py’)

Routes

- **‘/’**
  - Renders the **‘select_technique.html’** page where users can select a compression technique.
- **‘/select_technique’**
  - Handles POST requests from the technique selection form and renders **‘index.html’** with the selected technique.
- **‘/compress’**
  - Handles image upload and compression. It applies the selected compression technique (PCA, SVD, or ICA), calculates the PSNR, and returns the compressed image and PSNR value as a JSON response.

Compression Functions

- **‘apply_pca(channel, n_components)’**
  - Compresses the image using PCA.
- **‘apply_svd(channel, n_components)’**
  - Compresses the image using SVD.
- **‘apply_ica(channel, n_components)’**
  - Compresses the image using ICA.

PSNR Calculation

- **‘calculate_psnr(original_image, compressed_image)’**
  - Calculates the Peak Signal-to-Noise Ratio between the original and compressed images.

6.2. HTML Templates

‘select_technique.html’

- Provides buttons for selecting the compression technique (PCA, SVD, or ICA).
- Simple and clean interface with three buttons for technique selection.

‘index.html’

- Allows users to upload an image, view the original and compressed images, adjust the number of components via a slider, and download the compressed image.
- Contains JavaScript to handle image upload, display previews, and communicate with the Flask backend to perform compression and display results.
- User-friendly design with a focus on ease of use.

JavaScript in ‘index.html’

- Handles file upload, image preview, component adjustment, and form submission for image compression.
- Displays a loading spinner while the compression is in progress.
- Updates the UI with the compressed image, PSNR value, and provides a download link for the compressed image.

7. Conclusion

The Image Compression Web Application provides a comprehensive platform for exploring advanced image compression techniques. By providing an intuitive interface and real-time feedback through PSNR values, the application allows users to understand the trade-offs between compression ratio and image quality. Its user-friendly interface and interactive features make it an excellent tool for educational purposes, allowing students and researchers to gain practical insights into image compression. Furthermore, its practical applications extend to professionals seeking efficient methods for reducing image file sizes while maintaining quality. The integration of PCA, SVD, and ICA within a single tool highlights the versatility and effectiveness of these techniques in the field of image processing.

8. Future Work

While the Image Compression Web Application offers a robust and interactive platform for experimenting with PCA, SVD, and ICA compression techniques, there are several areas for potential enhancement and expansion. Future work on this project could include the following:

1. **Integration of Additional Compression Techniques**: Incorporating other advanced compression methods such as wavelet transform or deep learning-based techniques to provide users with a broader range of options.
2. **User Interface Enhancement**: Improving the design and usability of the interface, including the addition of more interactive elements and better visualization tools.
3. **Use Advanced Quality Metrics**: Implementing additional quality metrics such as Structural Similarity Index (SSIM) or Visual Information Fidelity (VIF) for a more comprehensive evaluation of compression quality.

9. Reference

- Geeksforgeeks.org
- Github.com
- Flask.palletsprojects.com
- scikit-learn.org
- chatgpt.com